

Project 4

Title: Nonparametric statistics for evaluating censored reliability functions

Learning objectives: The overall learning objective of this project is to utilize focused analysis of reliability functions and to apply reliability functions to large datasets with only partially known values, commonly referred to as censored datasets. Specifically, the project will require students to apply estimators of survival functions to the provided dataset in order to evaluate the differential mortality of cancer patients harboring different types of molecular features. Students will need to consider the appropriate metadata type to which an estimator of a survival function should be applied to, while simultaneously accounting for any censoring present within the selected data type. Moreover, students will likely need to utilize nonparametric product limit estimators to evaluate and compare different survival functions amongst subpopulations of cancer patients. Lastly, students will also need to correct for confounding factors that may affect their results by leveraging appropriate statistical approaches, including, but not limited to, parametric survival or proportional hazards models.

Subject Matter: The focus in project 4 will be the construction of genome-wide survival models based on gene expression and somatic mutation data to identify likely prognostic clinical biomarkers in cancer patients. The project will require students to select the appropriate clinical endpoint for their specific cancer type and to consider the data structure of that endpoint, including any censoring of the data. Students will need to apply a non-parametric statistical approach, such as a Kaplan–Meier estimator, to the selected endpoint in order to evaluate the differential survival between different subsets of cancer patients partitioned based on either their gene expression and/or somatic mutation data. Additionally, students will also need to adjust for known covariates of the clinical endpoint by utilizing parametric survival and proportional hazards models, such as Cox proportional hazards model. Where applicable, these covariates should include age of diagnosis, sex, tumor stage, tumor grade, and any previously developed cancer-type specific molecular subtyping.

Project's manuscript: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9042322/>

Manuscript presentation: Team presentation on Tuesday (20-May) at 6:30pm (CENTR 217B)

Project's datasets: <https://www.dropbox.com/scl/fo/7d37xqur5v1b8jni61b0t/AC1f0-cbGkGBSZFI9fXzdQ?rlkey=pfw7xmb7slnz7d398gzfzpu7>

Project's presentation: Team presentation on Thursday (29-May) at 6:30pm (CENTR 217B)

Deadline: All teams submit a project report (max 3 pages) by 11:59pm on Friday (30-May)

Potential outline for approaching the project:

- Identify and justify which clinical endpoint (overall survival, disease-free survival, *etc.*) should be utilized for your dataset by considering your cancer type, the available data, and any potential censoring of these data. If you are unsure which endpoint is appropriate, you should compare your results for different clinical endpoints.
- Identify a set of gene expression and/or somatic mutations that should be tested for being prognostic biomarkers. As part of your selection, you should consider the prevalence of the selected features within the available data.
- Utilize a non-parametric statistical approach to compare the survival in patients with and without each selected prognostic biomarker.
- Correct for the known prognostic covariates specific for your data set by using either a parametric survival or utilizing a proportional hazards model.
- Be careful of multiple hypothesis testing issues and consider reducing the number of evaluated prognostic biomarkers.